c. Tarnishing of metal articles

Shiny metal surfaces and other articles lose their shining appearance due to chemical reactions on the surface. For example, silver articles become black when exposed to atmospheric air. Similarly, brass vessels which contain copper as one of the constituents develop a greenish layer when exposed to air for a long time. This is due to a chemical reaction between copper and moist air to form basic copper carbonate and copper hydroxide.



Fig 10.7 Damaged iron sheets

10.2.3 Production of Heat, Light, Sound and Pressure

a. Production of Heat

Have you ever rubbed your palms in winter season to keep yourself warm? Have you noticed the heat produced when you use cycle pump? Similarly some chemical reactions produce heat energy also. Such reactions are called **exothermic reactions**. For example, when you add water to quicklime (calcium oxide), lot of heat is released to produce slaked lime (calcium hydroxide).

Activity 4

Take two clean test tubes. Take sulphuric acid in one test tube and a solution of sodium hydroxide in another tube. Slowly and carefully add sodium hydroxide solution to sulphuric acid. Touch the sides of test tube. What do you feel? What do you infer?

b. Production of Light

When you ignite a candle, you get light as a result of burning. Some chemical reactions produce light. For example, when a piece of magnesium ribbon is burnt in a flame, bright light is produced with heat. Even the fireworks used during festival times produce different coloured lights which are all due to chemical reactions.

c. Production of Sound

When we speak sound is produced. When you hit metals like iron, copper etc., a sound is heard. Some chemical reactions do produce sound when they take place. What happens when you fire crackers during Deepavali? The chemical substances present in the crackers undergo some chemical reactions to produce sound.

Activity 5

Take a clean test tube. Add some dilute hydrochloric acid. Drop a piece of magnesium or a piece of zinc metal. What do you see? Now bring a burning match stick near the mouth of the test tube. What do you hear? What do you infer?

You can hear a pop sound. When metals like zinc or magnesium reacts with dilute acids hydrogen gas is produced. Since hydrogen gas is highly flammable it reacts with oxygen present in air to produce pop sound.

d. Production of Pressure

When you compress hard a balloon having full of air, it will burst. This is due to sudden release of air from the balloon as a result of increased pressure on compression. Some chemical reactions produce gases which increases the pressure when the reaction takes place in a closed container. If the pressure level goes beyond the limit, we get the explosion. Explosives and fireworks burst because of this reason. When they are ignited they explode due to pressure generated by gases from the chemical reactions. Thus, you hear a huge sound.

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